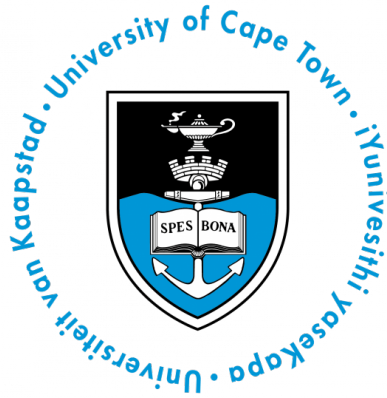




UNIVERSITY OF CAPE TOWN

MULTIVARIATE ANALYSIS

Continuous Assessment 9

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Question 1:

For the $p = 3$ standardized random variables Z_1, Z_2 and Z_3 :

$$Z_1 = 0.9F_1 + \epsilon_1$$

$$Z_2 = 0.7F_2 + \epsilon_2$$

$$Z_3 = 0.5F_3 + \epsilon_3$$

with $\text{Var}[F_1] = 1$, $\text{Cov}[\epsilon, F_1] = \mathbf{0}$ and

$$\Psi = \text{Cov}[\epsilon] = \begin{bmatrix} 0.19 & 0 & 0 \\ 0 & 0.51 & 0 \\ 0 & 0 & 0.75 \end{bmatrix}$$

We can write the covariance matrix in the following form:

$$\boldsymbol{\rho} = \mathbf{L}\mathbf{L}^T + \Psi$$

where

$$\mathbf{L} = \begin{bmatrix} 0.9 \\ 0.7 \\ 0.5 \end{bmatrix}$$

From the above

$$\begin{aligned} \mathbf{L}\mathbf{L}^T + \Psi &= \begin{bmatrix} 0.9 \\ 0.7 \\ 0.5 \end{bmatrix} \begin{bmatrix} 0.9 & 0.7 & 0.5 \end{bmatrix} + \begin{bmatrix} 0.19 & 0 & 0 \\ 0 & 0.51 & 0 \\ 0 & 0 & 0.75 \end{bmatrix} \\ &= \begin{bmatrix} 0.81 & 0.63 & 0.45 \\ 0.63 & 0.49 & 0.35 \\ 0.45 & 0.35 & 0.25 \end{bmatrix} + \begin{bmatrix} 0.19 & 0 & 0 \\ 0 & 0.51 & 0 \\ 0 & 0 & 0.75 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0.63 & 0.45 \\ 0.63 & 1 & 0.35 \\ 0.45 & 0.35 & 1 \end{bmatrix} \end{aligned}$$

Thus, the covariance matrix

$$\boldsymbol{\rho} = \begin{bmatrix} 1 & 0.63 & 0.45 \\ 0.63 & 1 & 0.35 \\ 0.45 & 0.35 & 1 \end{bmatrix}$$

Question 2:

A: Sample Correlation Matrix

	Wind	Radiation	CO	NO	NO2	O3	HC
Wind	1.00	-0.10	-0.19	-0.27	-0.11	-0.25	0.16
Radiation	-0.10	1.00	0.18	-0.07	0.12	0.32	0.05
CO	-0.19	0.18	1.00	0.50	0.56	0.41	0.17
NO	-0.27	-0.07	0.50	1.00	0.30	-0.13	0.23
NO2	-0.11	0.12	0.56	0.30	1.00	0.17	0.45
O3	-0.25	0.32	0.41	-0.13	0.17	1.00	0.15
HC	0.16	0.05	0.17	0.23	0.45	0.15	1.00

Table 1: Correlation Matrix of Environmental Variables

B: Principal Component Factor Model

The principal component method was used to obtain factor model solutions for $m = 1$ and $m = 2$ factors based on the sample correlation matrix.

Eigenvalues

Component	Eigenvalue
1	2.3368
2	1.3860
3	1.2041
4	0.7271
5	0.6535
6	0.5367
7	0.1559

Table 2: Eigenvalues of the Correlation Matrix

The first two principal components have eigenvalues greater than 1, suggesting that a two-factor model may be appropriate.

Factor Loadings for $m = 1$

Variable	Factor 1
Wind	0.3620
Radiation	-0.3142
CO	-0.8424
NO	-0.5772
NO2	-0.7613
O3	-0.4961
HC	-0.4883

Table 3: Factor Loadings ($m = 1$)

The proportion of total variance explained by the single factor is:

$$\frac{\lambda_1}{p} = \frac{2.3368}{7} = 0.3338$$

Factor Loadings for $m = 2$

Variable	Factor 1	Factor 2
Wind	0.3620	0.3278
Radiation	-0.3142	-0.6200
CO	-0.8424	-0.0080
NO	-0.5772	0.5117
NO2	-0.7613	0.2352
O3	-0.4961	-0.6675
HC	-0.4883	0.3625

Table 4: Factor Loadings ($m = 2$)

The proportion of variance explained by each factor is:

Factor 1: 0.3338, Factor 2: 0.1980

Cumulative: 0.5318

The two-factor model explains approximately 53.18% of the total variance, representing a substantial improvement over the single-factor model.

C: Maximum Likelihood Factor Model

A maximum likelihood factor analysis was performed with $m = 2$ factors using the `factanal()` function.

Factor Loadings

Variable	Factor 1	Factor 2
Wind	-0.1676	-0.2547
Radiation	0.0435	0.3206
CO	0.7826	0.4183
NO	0.6965	-0.1284
NO2	0.5965	0.1725
O3	-0.0082	0.9975
HC	0.2459	0.1559

Table 5: Maximum Likelihood Factor Loadings ($m = 2$)

Specific Variances (Uniquenesses)

Variable	Uniqueness
Wind	0.9070
Radiation	0.8953
CO	0.2126
NO	0.4984
NO2	0.6144
O3	0.0050
HC	0.9152

Table 6: Specific Variances (Ψ)

Variance Explained

Factor 1: 0.221, Factor 2: 0.201, Cumulative: 0.422

D: Comparison

A comparison was conducted between the principal component (PC) and maximum likelihood (ML) factor solutions for $m = 2$ factors.

Factor Loadings Comparison

Variable	PC Method		ML Method	
	F1	F2	F1	F2
Wind	0.3620	0.3278	-0.1676	-0.2547
Radiation	-0.3142	-0.6200	0.0435	0.3206
CO	-0.8424	-0.0080	0.7826	0.4183
NO	-0.5772	0.5117	0.6965	-0.1284
NO2	-0.7613	0.2352	0.5965	0.1725
O3	-0.4961	-0.6675	-0.0082	0.9975
HC	-0.4883	0.3625	0.2459	0.1559

Table 7: Comparison of Factor Loadings (PC vs ML)

While both methods identify similar groupings of variables, the magnitude and signs of the loadings differ due to the different estimation approaches. The ML method produces more interpretable structure, particularly with O_3 loading almost entirely on the second factor.

Variance Explained

- PC Method (cumulative): 53.18%
- ML Method (cumulative): 42.2%

The principal component method explains a larger proportion of total variance, as it is designed to maximise variance capture. In contrast, the ML method focuses on reproducing the covariance structure rather than maximising explained variance.

Model Fit Comparison

To assess model fit, the sum of squared off-diagonal residuals was computed:

- PC Method: 0.4985
- ML Method: 0.1995

The ML method yields substantially smaller residuals, indicating a better fit to the observed correlation structure. This is consistent with the objective of maximum likelihood estimation, which explicitly minimises discrepancies between the observed and model-implied covariance matrices.

E: Varimax Rotation and Interpretation

A varimax rotation was applied to the principal component solution with $m = 2$ factors to improve interpretability.

Rotated Factor Loadings

Variable	Factor 1	Factor 2
Wind	0.1236	0.4725
Radiation	0.0763	-0.6909
CO	-0.7007	-0.4677
NO	-0.7632	0.1124
NO2	-0.7659	-0.2198
O3	-0.0499	-0.8302
HC	-0.6070	0.0362

Table 8: Varimax Rotated Loadings

Variance Explained

Factor 1: 0.2931, Factor 2: 0.2387, Cumulative: 0.5318

Interpretation

The varimax rotation produces a clearer and more interpretable factor structure by concentrating high loadings on fewer variables.

- **Factor 1 (Pollution Factor):** Strong negative loadings on CO, NO, NO₂, and HC indicate that this factor captures overall pollution concentration. These variables now load predominantly on a single factor, improving interpretability.
- **Factor 2 (Atmospheric/Environmental Factor):** Strong loadings on O₃ and Radiation, with a moderate contribution from Wind, suggest that this factor represents atmospheric or environmental conditions influencing air quality.

F: Factor Scores using Weighted Least Squares

Factor scores for the $m = 2$ maximum likelihood model were estimated using the weighted least squares (WLS) approach.

Methodology

The factor scores are computed using the matrix formula:

$$\hat{\mathbf{F}} = (\mathbf{L}'\mathbf{\Psi}^{-1}\mathbf{L})^{-1}\mathbf{L}'\mathbf{\Psi}^{-1}(\mathbf{X} - \boldsymbol{\mu})$$

Since the analysis is based on the correlation matrix, the data are standardised, and thus $\boldsymbol{\mu} = \mathbf{0}$. The factor scores simplify to:

$$\hat{\mathbf{F}} = \mathbf{A}\mathbf{X}$$

where:

$$\mathbf{A} = (\mathbf{L}'\mathbf{\Psi}^{-1}\mathbf{L})^{-1}\mathbf{L}'\mathbf{\Psi}^{-1}$$

Coefficient Matrix

	Wind	Radiation	CO	NO	NO2	O3	HC
Factor 1	-0.0407	0.0107	0.8097	0.3075	0.2136	-0.3600	0.0591
Factor 2	-0.0014	0.0018	0.0098	-0.0013	0.0014	0.9969	0.0009

Table 9: WLS Coefficient Matrix **A**

Estimated Factor Scores

Observation	Factor 1	Factor 2
1	1.6787	-0.2303
2	0.1132	-0.7910
3	-0.2074	-0.6139
4	-0.2268	1.0055
5	-0.5405	0.1046
6	0.2733	0.4718
7	2.0135	1.0225
8	2.0022	0.8402
9	-0.0728	0.2923
10	0.5147	-0.0678

Table 10: First 10 Estimated Factor Scores

Interpretation

The factor scores represent the estimated values of the latent factors for each observation. Factor 1 primarily reflects overall pollution intensity, while Factor 2 captures ozone-related atmospheric conditions. Observations with high positive scores on Factor 1 correspond to higher levels of pollutants such as CO, NO, and NO₂, whereas high scores on Factor 2 indicate stronger influence from O₃. These scores can be used to compare observations in terms of their underlying factor structure.